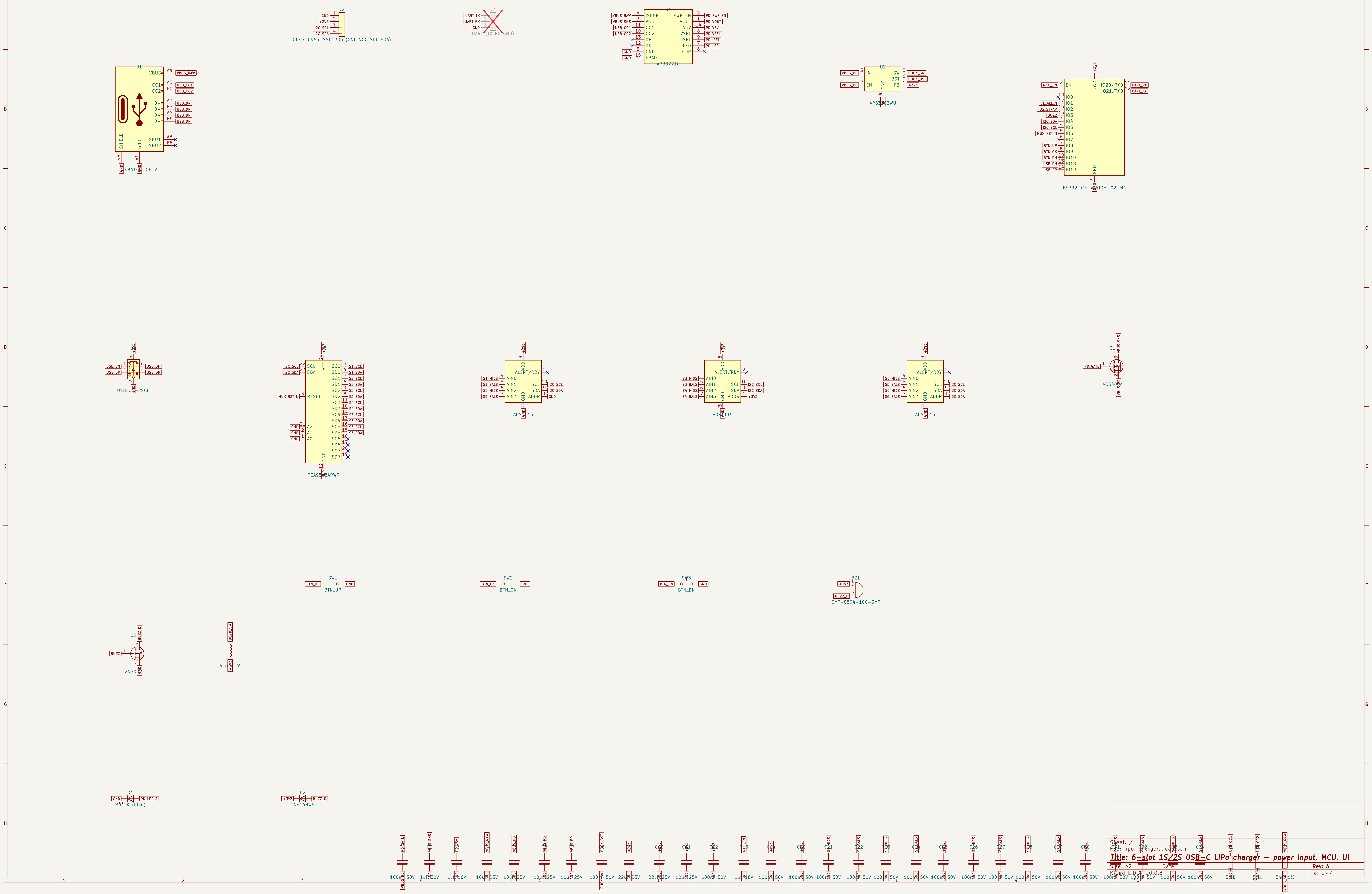
USB-C PD input (AP33771C asks for 9 V), 3.3 V buck, ESP32-C3, OLED + 3 buttons, I2C mux to the six slots.
Label-style schematic generated from design.py: pins with the same label are connected.

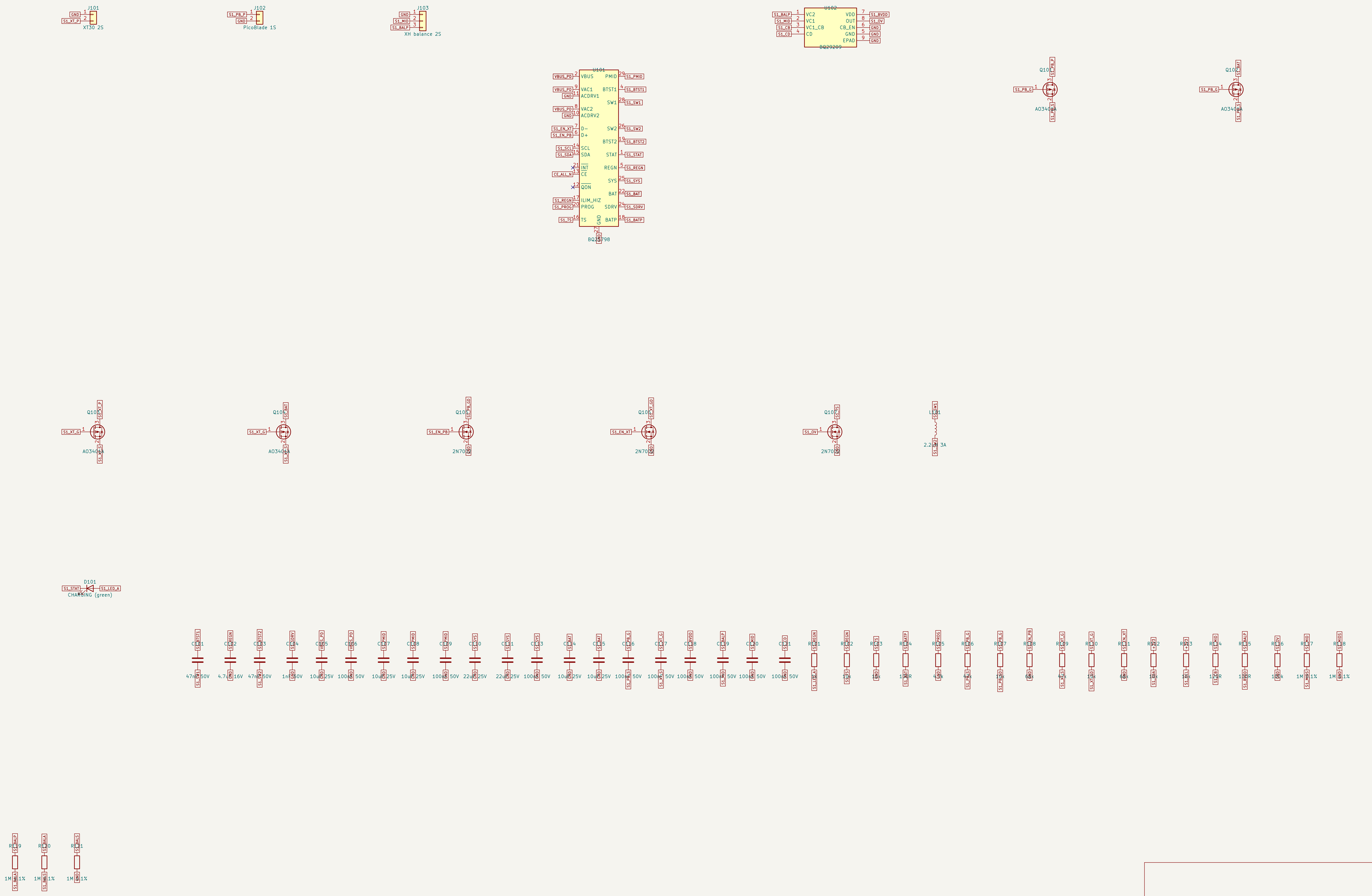


Sheet: /
File: lipo-charger.kicad_sch

Title: 6-slot 1S/2S USB-C LiPo charger - power input, MCU, UI

Size: A2
Date:
Rev: A
KiCad E.D.A. 10.0.8
Id: 1/7

BQ25798 buck-boost charger. D+ drives the PicoBlade switch, D− the XT30 switch (register 0x47).
Each connector has a back-to-back P-FET pair so the two packs can never be connected together.

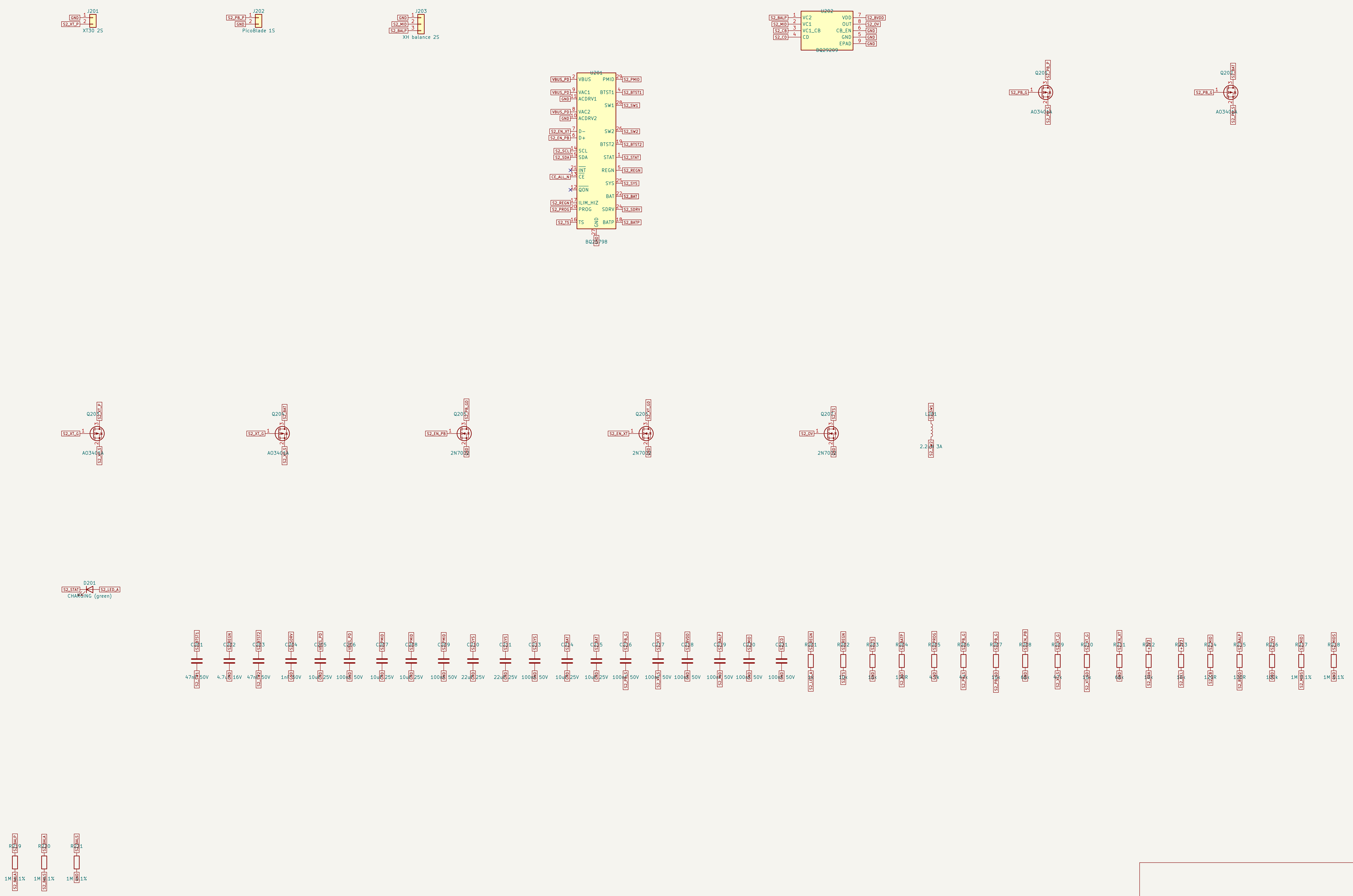


Sheet: /Slot 1/
File: slot1.kicad_sch

Title: Charger slot 1: BQ25798 + PicoBlade (1S) / XT30 (2S) switches

Size: A2
Date:
Rev: A
KiCad E.D.A. 10.0.8
Id: 2/7

BQ25798 buck–boost charger. D+ drives the PicoBlade switch, D– the XT30 switch (register 0x47).
Each connector has a back–to–back P–FET pair so the two packs can never be connected together.

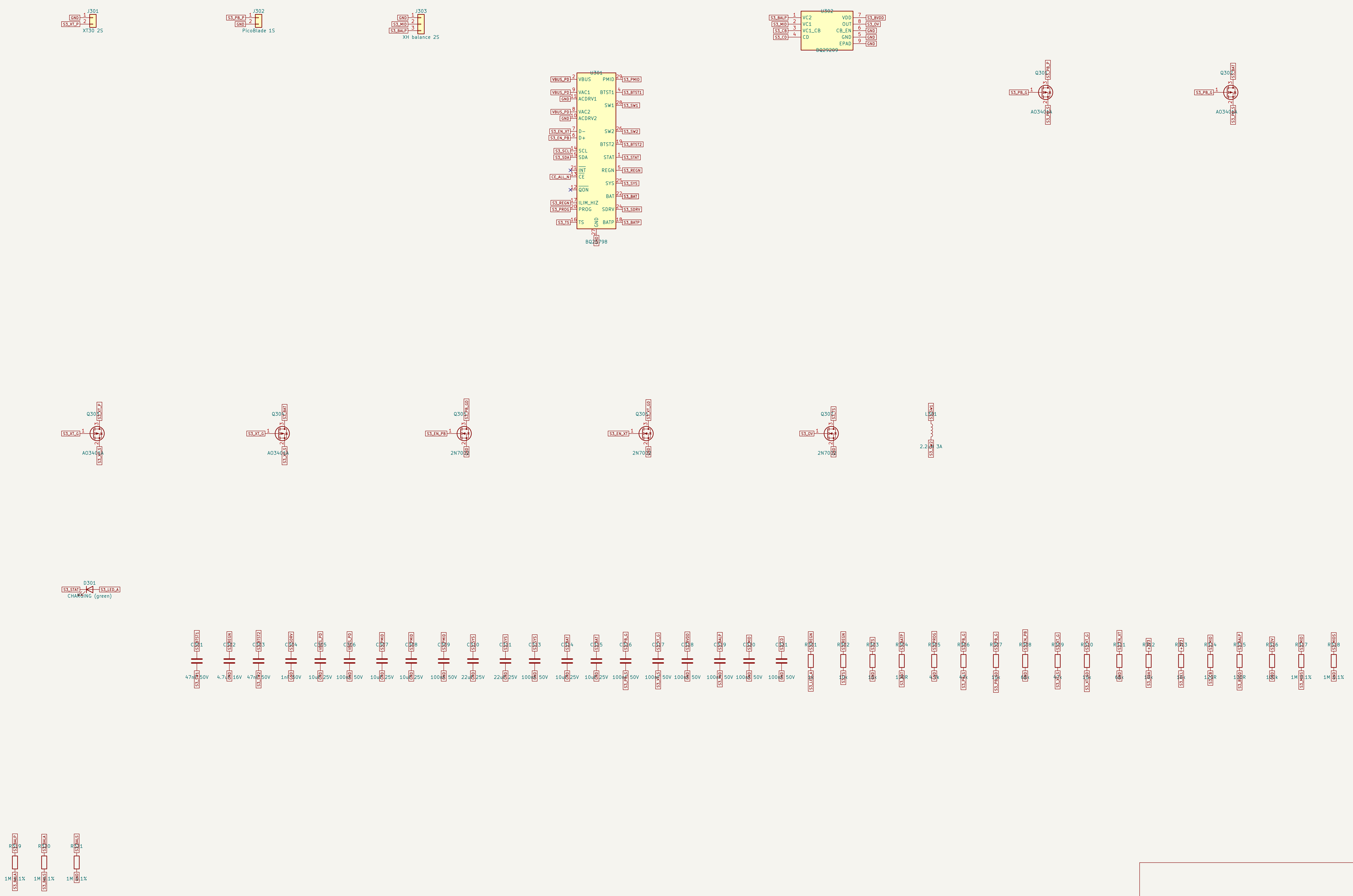


Sheet: /Slot 2/
File: slot2.kicad_sch

Title: Charger slot 2: BQ25798 + PicoBlade (1S) / XT30 (2S) switches

Size: A2 Date:
Rev: A
KiCad E.D.A. 10.0.8
Id: 3/7

BQ25798 buck-boost charger. D+ drives the PicoBlade switch, D− the XT30 switch (register 0x47).
Each connector has a back-to-back P-FET pair so the two packs can never be connected together.

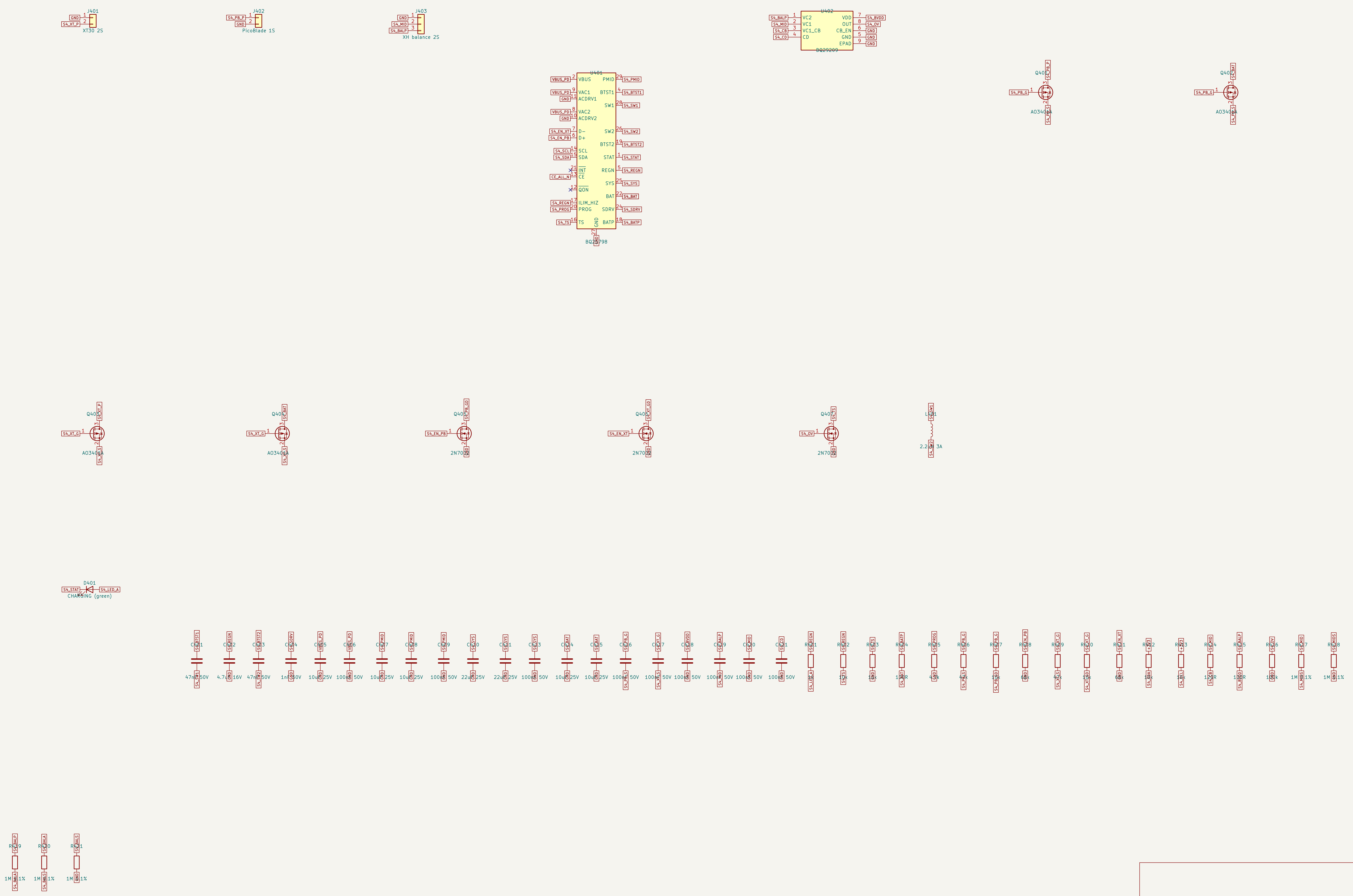


Sheet: /Slot 3/
File: slot3.kicad_sch

Title: Charger slot 3: BQ25798 + PicoBlade (1S) / XT30 (2S) switches

Size: A2
Date:
Rev: A
KiCad E.D.A. 10.0.8
Id: 4/7

BQ25798 buck-boost charger. D+ drives the PicoBlade switch, D− the XT30 switch (register 0x47).
Each connector has a back-to-back P-FET pair so the two packs can never be connected together.

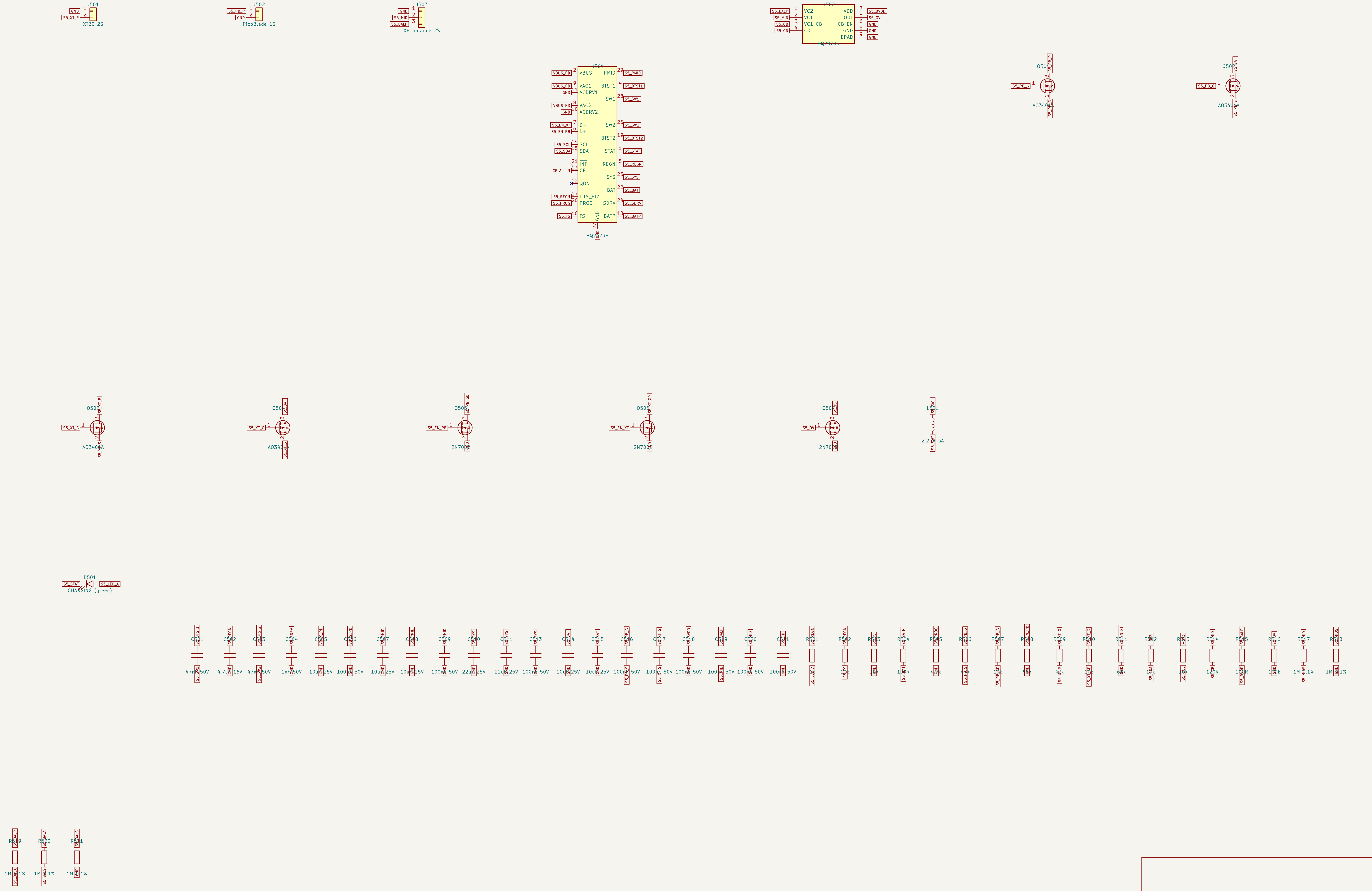


Sheet: /Slot 4/
File: slot4.kicad_sch

Title: Charger slot 4: BQ25798 + PicoBlade (1S) / XT30 (2S) switches

Size: A2 Date: Rev: A
KiCad E.D.A. 10.0.8 Id: 5/7

BQ25798 buck–boost charger. D+ drives the PicoBlade switch, D– the XT30 switch (register 0x47).
Each connector has a back–to–back P–FET pair so the two packs can never be connected together.

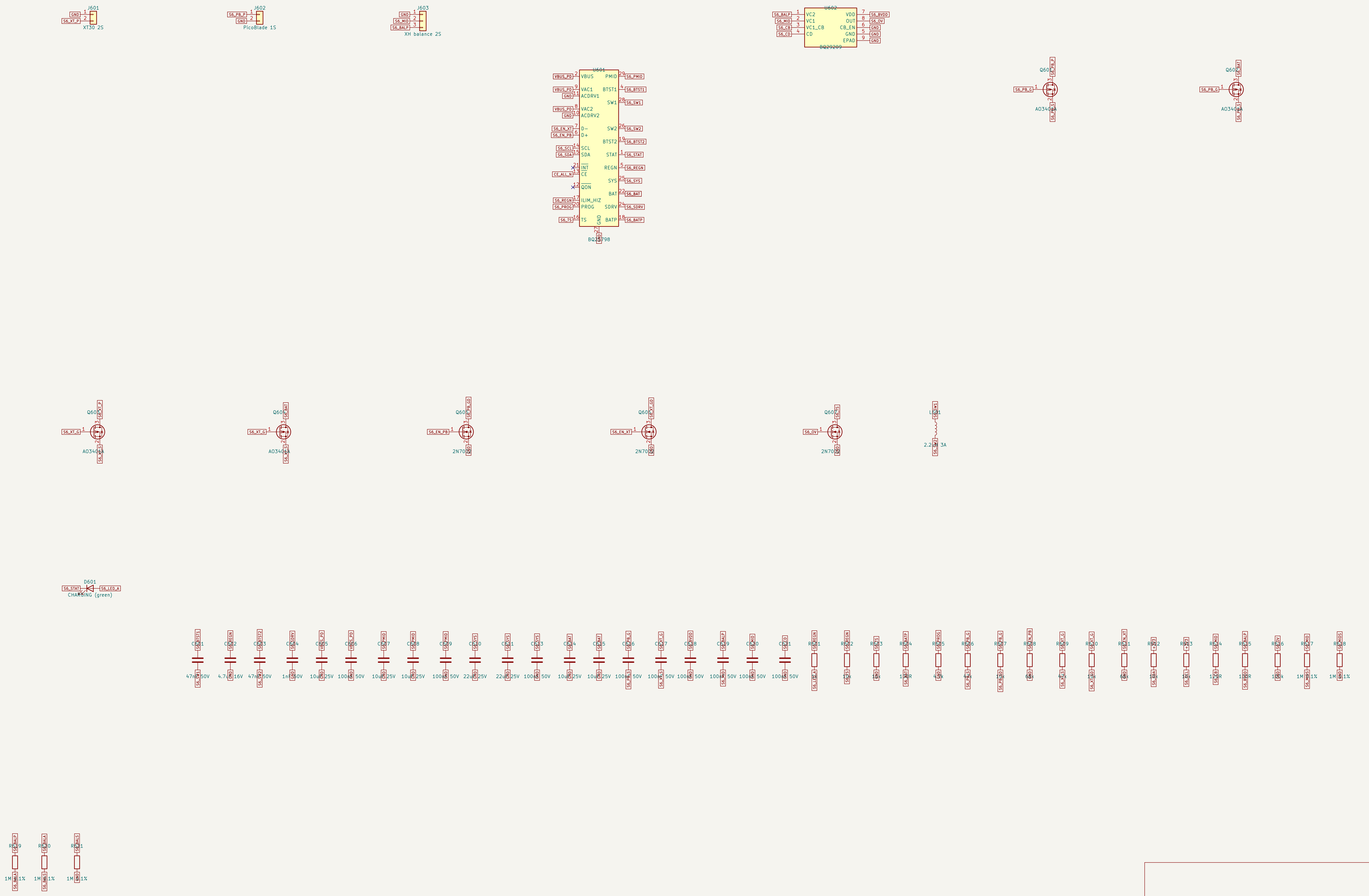


Sheet: /Slot 5/
File: slot5.kicad_sch

Title: Charger slot 5: BQ25798 + PicoBlade (1S) / XT30 (2S) switches

Size: A2
Date:
Rev: A
KiCad E.D.A. 10.0.8
Id: 6/7

BQ25798 buck-boost charger. D+ drives the PicoBlade switch, D- the XT30 switch (register 0x47).
Each connector has a back-to-back P-FET pair so the two packs can never be connected together.



Sheet: /Slot 6/
File: slot6.kicad_sch

Title: Charger slot 6: BQ25798 + PicoBlade (1S) / XT30 (2S) switches

Size: A2
Date:
Rev: A
KiCad E.D.A. 10.0.8
Id: 7/7